

Chapter 1

Fundamentals of Artificial Intelligence [SEng9141]

Learning Objectives

By the end of this module, students will be able to:

- Define Artificial Intelligence in multiple ways.
- Distinguish between the four main historical approaches to AI: Thinking Humanly, Acting Humanly, Thinking Rationally, Acting Rationally.
- Articulate the main goals and foundational disciplines of AI.
- Recall key milestones in the history of AI.
- Identify prominent examples of modern AI applications in the real world.

Basic outline

- ✓ Introduction to AI Objectives/Goals of AI What is AI?
- ✓ Approaches to AI – making computer:
 - ✓ Think like a human (Thinking humanly)
 - ✓ Act like a human (Acting humanly)
 - ✓ Think rationally (Thinking rationally)
 - ✓ Act rationally (Acting rationally)
 - ✓ The Foundations of AI
 - ✓ Bits of History and the State of the Art

1.1 introduction

What is AI?

- ❖ Artificial intelligence (AI) is technology that enables computers and machines to simulate human learning, comprehension, problem solving, decision making, creativity and autonomy.
- ❖ Applications and devices equipped with AI **can see** and identify objects.
- ❖ They can understand and respond to human language.
- ❖ They can learn from new information and experience.
- ❖ They can make detailed recommendations to users and experts.
- ❖ They can act independently, replacing the need for human intelligence or intervention (a classic example being a self-driving car).

Key Objectives and Goals of AI

- **Simulate Human Intelligence:** Replicate human cognitive abilities (thinking, learning, creating) in machines.
- **Automate Complex Tasks:** Perform tasks like medical diagnosis, data analysis, or driving with greater accuracy and speed than humans.
- **Enhance Decision-Making:** Use data-driven insights to make better, faster, or more efficient decisions, such as in finance,, logistics, or AI-driven marketing.
- **Improve Productivity & Efficiency:** Reduce human workload by handling repetitive or complex tasks.
- **Adaptability & Learning:** Develop systems that can learn from new information or experience without being explicitly reprogrammed for every scenario.
- **Understanding Language & Perception:** Enabling machines to understand, interpret, and respond to human language and visual inputs.

Objectives of AI

Objectives describe what AI is designed to do in general the *core abilities* AI systems should have.

They focus on capabilities and functions.

Examples (Objectives):

- Simulate human intelligence
- Learn from data
- Reason and make decisions
- Understand natural language
- Recognize images and speech

Goals of AI

Goals describe why AI is developed the *purpose or outcomes* we want to achieve by using AI.

They focus on results and impact.

Examples (Goals):

- Automate tasks
- Improve efficiency and accuracy
- Reduce human effort and error
- Support human decision-making
- Solve real-world problems

Simple Comparison

Aspect	Objectives of AI	Goals of AI
Meaning	What AI can do	Why AI is used
Focus	Capabilities	Outcomes
Nature	Technical	Practical
Example	Learning, reasoning	Automation, efficiency

Approaches to AI

- ✓ Approaches to Artificial Intelligence (AI) for creating intelligent computers involve various methods.
- ✓ primarily driven by Machine Learning (ML), which enables systems to learn from data without explicit programming.
- ✓ Key approaches include
 - ✓ top-down (symbolic) AI for rule-based reasoning,
 - ✓ bottom-up (connectionist/deep learning) approaches for neural network pattern recognition, and
 - ✓ hybrid systems.

Key Approaches to Making AI Computers:

1) Machine Learning (ML): Computers identify patterns and relationships within data, improving performance over time.

- **Supervised Learning:** Training models on labeled data to map inputs to known outputs.
- **Unsupervised Learning:** Identifying hidden patterns or structures in unlabeled data.
- **Reinforcement Learning:** Systems learn by trial-and-error to maximize rewards.

Key Approaches to Making AI Computers...

- 2) **Deep Learning:** A subset of machine learning using neural networks (inspired by the human brain) to handle complex, large-scale data like image recognition and natural language processing (NLP).
- 3) **Symbolic (Top-Down) AI:** Known as "Good Old-Fashioned AI" (GOFAI), this approach uses explicit rules and logic to simulator human reasoning.
- 4) **Hybrid Learning:** Combining techniques (e.g., supervised + unsupervised) to leverage strengths and improve accuracy.
- 5) **Expert Systems:** Programs that mimic the decision-making abilities of a human expert in specific fields.

Approaches to AI – Making Computers:

- AI approaches focus on how machines mimic or achieve intelligence. Russell and Norvig categorize them into four:

1. Think Like a Human (Thinking Humanly):

- 1.Focus: Cognitive modeling simulate human thought processes.
- 2.Method: Introspection, psychological experiments, brain imaging.
- 3.Example: General Problem Solver (GPS) by Newell and Simon (1950s), which mimics human problem-solving steps.
- 4.Strengths: Helps understand human cognition.
- 5.Weaknesses: Human thinking is complex and not always optimal.

2 Act Like a Human (Acting Humanly):

- Focus: Behavior **imitation** pass the Turing Test (1950), where a machine fools a human into thinking it's human via conversation.
- Method: Natural language processing, knowledge representation, learning.
- Example: Chatbots like ELIZA (1960s) or modern ones like GPT models.
- Strengths: Useful for human-machine interaction.
- Weaknesses: Doesn't require true understanding; can be superficial.

3) Think Rationally (Thinking Rationally):

- Focus: Logical reasoning use laws of thought (e.g., syllogisms from Aristotle).
- Method: Formal logic, deduction, reading.
- Example: Expert systems that use rules to diagnose diseases.
- Strengths: Provably correct reasoning.
- Weaknesses: Real-world uncertainty and computational limits (e.g., undecidability).

Example: Thinking Rationally: "If road is wet, then slippery."

4) Act Rationally (Acting Rationally)

1. Focus: Rational agents act to achieve best outcomes.
 2. Method: Goal-oriented decision-making under uncertainty.
 3. Example: Autonomous vehicles optimizing for safety and efficiency.
 4. Strengths: Flexible and practical.
 5. Weaknesses: Requires defining "rationality" precisely.
- These approaches are not mutually exclusive; modern AI often combines them.

Example: Acting Rationally: "Camera sees wet road → reduce speed by 20% → check **traction** → adjust."

The Four Approaches To Ai Summary

Approach	Goal	Test/Measure	Example
Thinking Humanly	Model human thought	Match brain/thinking patterns	Cognitive science models, brain scans
Acting Humanly	Mimic human behavior	Turing Test	Chatbots, voice assistants
Thinking Rationally	Use perfect logic	Prove theorems correctly	Expert systems, logic programming
Acting Rationally	Achieve best outcomes	Maximize utility/performance	Self-driving cars, recommendation systems

Key Steps in Building an AI System:

- **Hardware Selection:** Utilizing GPUs for high-performance deep learning, efficient CPUs, and sufficient RAM.
- **Software Frameworks:** Employing tools like Python, TensorFlow, or PyTorch to develop models.
- **Data Acquisition:** Gathering and preparing large datasets for training.
- **Model Training and Optimization:** Refining algorithms to improve accuracy in prediction or generation tasks.

Functional Approaches (What the AI does):

- **Computer Vision:** Interpreting visual information.
- **Natural Language Processing (NLP):** Understanding and generating human language.
- **Robotics:** Enabling autonomous physical actions.
- **Predictive Analytics:** Forecasting future outcomes based on historical data.

What Do You Think?

- Is Artificial Intelligence Good for Society?
- Explore the ProCon debate

The Foundations of AI

AI draws from:

- **Philosophy:** Questions of mind, knowledge (e.g., Descartes' dualism).
- **Mathematics:** Logic, probability, algorithms (e.g., Bayes' theorem for uncertainty).
- **Economics:** Decision theory, utility maximization.
- **Neuroscience:** Brain-inspired models (e.g., neural networks).
- **Psychology:** Behaviorism and cognitive science.
- **Computer Engineering:** Hardware for computation.
- **Linguistics:** Language understanding.

Knowledge Representation in AI

- Knowledge representation (KR) in AI **is** the method of encoding, organizing, and structuring information into machine-understandable formats.
- such as logical rules, semantic networks, frames, and neural network weights.
 - to enable AI systems
 - to perform reasoning, learning, and decision-making.
- It bridges the gap between raw data and actionable intelligence, allowing machines to simulate human cognitive abilities

Key Knowledge Representation Techniques

- **Logical Representation:** Uses formal, symbolic logic (like first-order logic) to represent facts and rules, allowing the system to deduce new information.
- **Semantic Networks:** Represents knowledge as a graph of interconnected nodes (concepts) and edges (relationships), such as "Dog $\rightarrow$ Is-A $\rightarrow$ Animal".
- **Frames:** Structured data representations that describe stereotyped situations or objects with slots, attributes, and default values.
- **Production Rules:** Uses "IF-THEN" statements to encode knowledge and make inferences (e.g., IF it is raining, THEN take an umbrella).

Cont...

- **Ontologies:** Provides a formal, structured, and explicit specification of concepts, properties, and relationships within a specific domain to ensure consistency.
- **Neural Networks/Embeddings:** In machine learning, knowledge is often represented implicitly as patterns of weighted connections between nodes (vectors).

Types of Knowledge Represented

- **Declarative:** Facts, truths, and "what" is known.
- **Procedural:** "How-to" knowledge regarding steps to perform tasks.
- **Heuristic:** Rules of thumb or *experiential* knowledge.
- **Meta-knowledge:** Knowledge about knowledge (structure, validity).
- The goal of these representations is to make knowledge accessible for automated reasoning, allowing AI to solve complex, real-life problems.

Bits of History and the State of the Art

- **Early History:** 1940s-1950s: McCulloch-Pitts neuron model; Turing's work; Dartmouth Conference (1956) coins "AI."
- **Boom and Bust:** 1960s-1970s: Early successes (e.g., SHRDLU for language); AI winters due to overpromises.
- **1980s-1990s:** Expert systems boom; machine learning rises.
- **Current State (as of 2026):** Deep learning dominates (e.g., transformers in NLP); applications in self-driving cars, recommendation systems, medical diagnostics. Challenges: Ethics, bias, energy consumption.

Common Misconceptions

- Myth 1: "AI = Robots"
- Truth: Most AI is software, not robots
- Myth 2: "AI will become conscious"
- Truth: Current AI has no consciousness or feelings
- Myth 3: "AI understands like humans"
- Truth: AI finds patterns in data; doesn't "understand"
- Myth 4: "AI is infallible"
- Truth: AI makes mistakes, often in surprising ways

Cont..

- ✗ "AI = Robots" → Most AI is software, not physical robots
- ✗ "AI understands like humans" → AI finds patterns, doesn't truly "understand"
- ✗ "AI is perfect" → AI makes mistakes, often in surprising ways
- ✗ "AI has feelings" → Current AI has no consciousness or emotions
- ✗ "AI will take over the world soon" → General AI is still far away

Summary

1. AI has multiple definitions : choose based on goal
2. Four main approaches: Thinking/Acting × Humanly/Rationally
3. Modern AI focuses on rational agents
4. AI is interdisciplinary : draws from philosophy, math, psychology, etc.
5. History shows cycles of **hype** and disappointment
6. Today: Narrow AI is everywhere; AGI is the frontier

Lab set up

- Install VS Code (Recommended Editor)
- Download: <https://code.visualstudio.com/>
- Install Python Extension:
- Open VS Code
- Go to Extensions (Ctrl+Shift+X)
- Search for "Python" by Microsoft
- Install it
- Set Python Interpreter:
- Open Command Palette (Ctrl+Shift+P)
- Type: "Python: Select Interpreter"
- Choose: Python 3.10.x ('ai_course': conda)

SIMPLE EXERCISE

- "YOUR FIRST AI PROGRAM" Exercise 1: The Temperature Advisor
- Problem: Create a simple "intelligent" system that gives advice based on temperature.

This is all about chapter 1